

Go, as in Google

Named from the first two letters of Google itself, Go is aimed to be a programming language of tomorrow's concurrent, networked computing world

Go on Go?

The Google Go web site golang.org runs on the godoc documentation server which is written in Go itself



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{T} here is a reason Google appeals to geeks; they give us so much fun stuff to play with, and often for free. Google Go is one of their latest new projects, born during the famous 20 per cent time that all Google employees use to develop innovative products on their own.

What is Go?

Go is a new open-source programming language created by Google which it claims to be simple, fast, safe, concurrent, fun. For someone who is familiar with programming languages such as Java and C++, it does not significantly depart from the syntax. The same familiar curly braces are there along with semicolons as line terminators. Yet there are enough idiosyncrasies to make the language new.

The language draws heavily from the basic C syntax, while bringing in some of the best features of other programming languages. It supports concurrency in its basic syntax, and has built-in garbage collection.

Why another language?

Google created Go since they believe that current programming languages are

not very suitable for creating the kind of applications that are needed nowadays.

Their issues with the state of affairs with developing in current programming languages is that despite computers being enormously faster now, software development has not speeded up much. Languages such as C and C++ use headers which make dependency analysis and fast compilation difficult.

Many of current programming languages also have a very "cumbersome" type system which pushes more people towards languages like Python and JavaScript which have dynamic typing. Most of them also have very limited features when it comes to parallelism and support

Google Go: Google's take on what a modern programming language should be like

a complicated syntax for the same.

Programming languages have kept on adding features via newer and newer libraries which bring new features, however the languages themselves have not evolved to take on the features which are required at their very core. Current languages have become very verbose requiring much "clerical work" in adding any functionality.

The shape of computing has changed significantly over the last couple of years,

and it is time for a change. With the introduction of a new programming language which is developed keeping the requirements of today in mind, developer and users alike have a lot to benefit for code which runs faster and is easier to develop.

Some important aspects

C++ is definitely one of the most popular languages for system development. Let us look at some of the ways in which Go differs from C++, since many of the points have already been mentioned before, they will be given in brief:

- No pointer arithmetic
- No type hierarchy
- No overloading methods or operators
- No implicit type conversions
- No header files, Go uses the concept of packages instead
- Strings, arrays and maps (hashes) are a native type in Go
- Threading and concurrent programming features are natively provided by the language
- Variables are initiated on declaration, there are never any uninitiated variables.
- Internal garbage collection

The syntax for declaring variables is reversed, as:

```
var variable_name int;
var another_variable_name
```

string;

Instead of C++'s:

```
int variable_name;
char *another_variable_name;
```

• Functions are declared using the func keyword with the following syntax:

```
func some_function(a int, b
int, c string);
```

• If a variable is being assigned at the same time as it is being declared, the type may be omitted as follows:

```
var some_variable = some_
other_variable;
```

• There is no distinction in syntax for accessing properties of pointers to structure and structure, both are accessed via the . notation. This is unlike C++ where pointers to structures require a -> notation.

• Go supports multiple parallel assignments, as such swapping two variables can be written quite simply as follows:

```
i, j = j, i;
```